

INTERFACING SENSOR WITH ADC OF AVR MICROCONTROLLER

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Sensors mimic our biological senses and respond to physical stimuli such as light, heat or movement. These can sometimes even outperform our natural sense organs in certain conditions. One can have additional senses with direction sensors, gyroscopes and infrared sensors. The evolution of microelectromechanical systems (MEMS) sensor technology has surpassed our imagination through its applications in handheld electronic gadgets including digital cameras, cellphones, gaming stations to that in space robots, aircrafts, simulators, etc.

This article explains interfacing a sensor with AVR microcontroller and displaying the sensor's output on a PC monitor.

Sensor signal conversion

Normally a sensor consists of a sensing element and a transducer that

converts the energy into a desired form (normally electrical). For example, in a microphone the small displacement of the diaphragm is converted into electrical signals. Most of the sensing elements can be made to generate an analogue direct current (DC) voltage by suitable circuitry. Also, a simple photo-resistive sensor can be used to generate an equivalent voltage signal that can sense light intensity. Similarly, a negative temperature coefficient (NTC) thermistor or a centigrade temperature sensor can be used to convert heat to voltage signals.

Most of the sensors change their resistance or capacitance with change in physical parameters. The change in resistance or capacitance can be converted to voltage signals. Microcontroller simplifies the sensor circuitry by allowing voltage input (0-5V range) through its analogue-to-digital converter (ADC) pins, which can be programmed to trigger a function at



a particular value, or can produce variable functions at different values of input voltage.

In context to AVR microcontrollers, any sensed parameter can be fed to one of its single-ended ADC input channel, which accepts a positive DC voltage and converts the signal to equivalent 10-bit digital value in its ADC registers—ADCH and ADCL. The value in ADC registers is given by:

$$ADC = \frac{V_{IN} \times 1024}{V_{REF}}$$

V_{REF} is the selected reference voltage, which ensures the maximum value close to 1024 (3FF hexadecimal, 10-bit maximum value). Default value can be set at 2.56V internal supply.

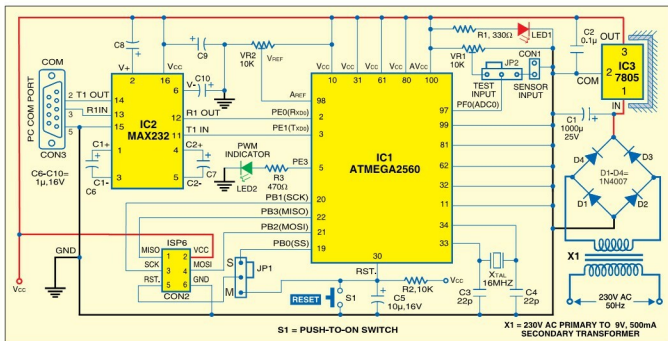


Fig. 1: Interfacing sensor with ADC of AVR microcontroller